FBSNet: Rethinking Directional Structure Modeling in Remote Sensing Segmentation via Frequency-Band State Space Networks

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***Abstract*—In remote sensing semantic segmentation, man-made objects often exhibit strong directional structures that manifest multifrequency cues ranging from low-frequency contextual layouts to high-frequency boundary details. However, most existing approaches either prioritize spatial-domain modeling or treat frequency cues only as auxiliary enhancements and fusion signals, leaving hierarchical frequency organization and interband dependencies underexplored. To address this, we propose FBSNet, the first frequency-band state space network for remote sensing semantic segmentation. The key insight of FBSNet is to shift dependency modeling from the spatial domain to an ordered frequency-band sequence, enabling explicit cross-band coordination from low-frequency context to high-frequency directional details. First, its core frequency-band state space (FBSS) module learns a structured frequency-band decomposition and serializes sub-bands into a direction-aligned sequence, along which state space modeling captures cross-band dependencies and preserves orientation-consistent cues from context to boundary details. Second, we integrate a hierarchical spatial mamba (HSM) block into the decoder to enforce cross-scale structural continuity, yielding a balanced frequency–spatial representation. Moreover, a Fourier-guided refinement strategy leverages frequency-domain cues to mitigate boundary information loss during upsampling. Extensive experiments on ISPRS Vaihingen, ISPRS Potsdam, and LoveDA consistently validate the effectiveness of FBSNet, which attains state-of-the-art mIoU (85.84%/87.90%/55.47%). The code is available at: https://github.com/liye1213/FBSNet**

***Index Terms*—Directional structures, frequency band, mamba, semantic segmentation, state space model (SSM).**

I. Introduction

REMOTE sensing semantic segmentation is a fundamental task for interpreting high-resolution aerial and satellite imagery [1], serving critical applications such as urban planning, land resource monitoring [2], and disaster assessment [3], [4]. In such complex scenarios, man-made objects display inherent anisotropy and multiscale directional geometry, ranging from macroscale road infrastructures and strip-shaped farmlands to microscale-aligned vehicles [5], [6]. Despite significant progress in deep learning, existing methods often struggle to preserve the integrity of these structures, leading to connectivity breaks in elongated targets and blurred boundaries in dense urban areas [7]. Physically, these degradation issues stem from the insufficient coupling of multifrequency cues: low-frequency components encode the coarse structural layout and long-range continuity [8], while direction-selective high-frequency bands determine local oriented boundaries [9]. This intrinsic property suggests that a structured frequency-band representation could serve as a more effective intermediate for robust segmentation.

However, the mainstream paradigm in remote sensing segmentation still primarily focuses on spatial-domain modeling. While frequency cues are occasionally introduced, they are typically treated as auxiliary supplements for feature fusion rather than primary carriers of representation [10]. Consequently, the hierarchical coordination between the global layout (low frequency) and oriented fine details (high frequency) remains underexplored, motivating the need for explicit cross-band dependency modeling [11].

Spatial-domain representation learning has long dominated this field. Early convolutional neural network (CNN)-based approaches, such as U-Net [12] and PSPNet [13], established the baseline for multiscale feature aggregation. More recently, Transformers (e.g., Swin Transformer [14] and SegFormer [15]) and hybrid architectures like UNetFormer [16] have significantly strengthened long-range context modeling. Despite their success, these methods operate on spatial tokens within the image plane. This inherent spatial-grid nature hinders the explicit disentanglement of directional structures, which are more naturally organized within a hierarchical frequency-band space.

Parallel to spatial modeling, frequency-domain analysis offers a direct path to decouple coarse structures from details. Classical tools like the 2-D discrete wavelet transform (DWT) decompose features into approximation (LL) and directional details (LH, HL, and HH) [17]. Recent works, such as SFFNet [18] and MSFFNet [19], have adopted such decompositions for multiband representation. Others, like AFENet [20] and FGNet [21], leverage fast Fourier transform (FFT)-based adaptive enhancement for global–local integration. Nevertheless, a limitation persists: most existing methods rely on fixed bases (e.g., Haar or standard Fourier bases) or predefined partitions, which may not optimally align with the complex, varying textures of remote sensing imagery. Furthermore, frequency

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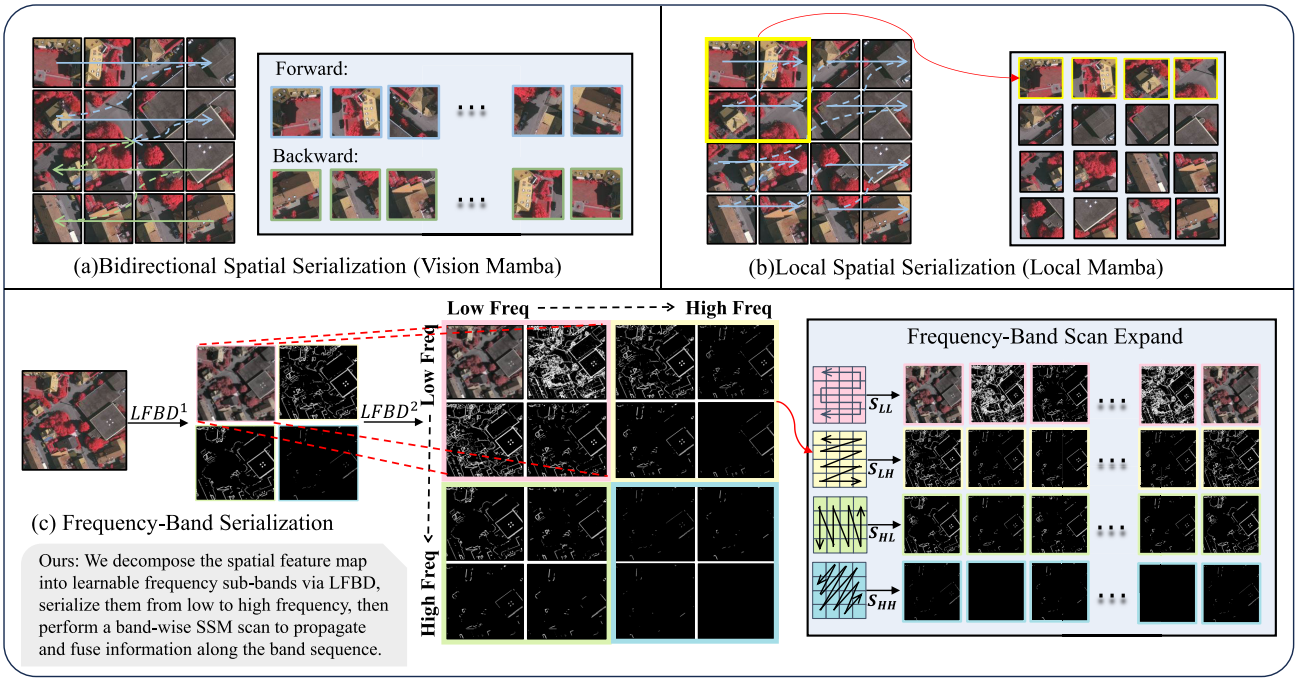


Fig. 1. Serialization axes in SSMs. (a) Bidirectional spatial scans (e.g., Vision Mamba [22]). (b) Local windowed spatial scans (e.g., LocalMamba [23]). (c) Frequency-band serialization (ours): LFBD decomposes features into ordered directional sub-bands (LL for coarse structure; LH/HL/HH for horizontal/vertical/diagonal details), and state propagation along the band sequence explicitly models cross-band dependencies for improved directional structure and boundary recovery.

components are usually injected into a spatial backbone via simple summation or concatenation, lacking a mechanism to explicitly model the sequential dependencies evolving from low-frequency context to high-frequency details.

Recently, state space models (SSMs), represented by VMamba [24], have emerged as a linear-complexity alternative for long-range dependency modeling and have been adapted for remote sensing (e.g., RS3Mamba [25] and CPSSNet [26]). The core of visual SSMs lies in serialization. Existing methods typically flatten 2-D feature maps into spatial sequences using bidirectional [Fig. 1(a)] or windowed [Fig. 1(b)] scan routes. While effective for spatial aggregation, these spatial-domain scans are not explicitly aligned with the hierarchical organization of directional frequencies. Multiresolution frequency decomposition (e.g., wavelet packet transform [27]) inherently organizes representations from low-frequency global layout to high-frequency directional details, yielding a natural ordered band sequence for modeling [28]. Therefore, shifting the propagation axis from the image plane to a frequency-band sequence [Fig. 1(c)] provides a more physically interpretable carrier for structured cross-band modeling.

Based on this insight, we propose the frequency-band state space network (FBSNet). The core idea is to treat hierarchical frequency bands as primary carriers of representation and model their dependencies via state space propagation. Specifically, we propose the frequency-band state space (FBSS) module, which employs a learnable frequency-band decomposition (LFBD) to generate task-adaptive sub-bands, thereby overcoming the rigidity of fixed wavelet bases. These sub-bands are then serialized into a direction-aligned sequence [Fig. 1(c)], where state propagation explicitly coordinates low-frequency continuity with high-frequency directional cues. Unlike prior frequency-enhancement schemes, FBSS explicitly organizes sub-bands hierarchically and models cross-band dependencies. To achieve a holistic representation, we introduce a parallel hierarchical spatial mamba (HSM) branch. Complementing the frequency branch, HSM enforces large-area geometric coherence in the spatial domain, ensuring that the sharpened directional details maintain global spatial consistency. Finally, a lightweight Fourier-guided refinement module (FGRM) is employed to preserve boundary fidelity during decoding.

The main contributions of this work are summarized as follows.

1) We present the first FBSS framework for directional structure modeling in remote sensing segmentation. It leverages cross-band dependencies to couple global layout with fine-grained directional details, enhancing directional structure modeling and contour extraction.
2) We introduce the HSM block to complement the frequency branch. By enforcing global geometric continuity and cross-scale spatial consistency, it works in synergy with FBSS to robustly handle complex urban–rural transitions and mitigate intraclass fragmentation.
3) We incorporate a lightweight FGRM in the decoder, which leverages global spectral priors to rectify high-frequency attenuation during upsampling, ensuring precise boundary delineation.
4) Extensive experiments on standard benchmarks demonstrate that FBSNet achieves state-of-the-art mIoU, effectively preserving directional structures and

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sharp boundaries with a favorable accuracy—efficiency trade-off.

II. Related Work

A. CNN- and Transformer-Based Methods

Spatial-domain representation learning has long dominated remote sensing image semantic segmentation. Early CNN-based frameworks, such as FCN by Long et al. [29] and the encoder–decoder U-Net by Ronneberger et al. [12], established the standard paradigm of dense prediction with multilevel feature fusion. Subsequent studies improved multiscale context aggregation and boundary quality via pyramid pooling and atrous convolution (e.g., PSPNet [13] and DeepLabv3 + [30]), while feature pyramids (e.g., FPN [31]) further enhanced robustness to scale variations in high-resolution imagery. To overcome the limited receptive field of CNNs, Transformer-based models were introduced to explicitly model long-range interactions. Dosovitskiy et al. [32] pioneered ViT by representing images as patch sequences and applying standard Transformer encoders to visual recognition. Hierarchical backbones (e.g., Swin Transformer) were further developed to better accommodate high-resolution imagery [14]. In RSI segmentation, representative Transformer-based designs include UNetFormer and DWin-HRFormer, which explore balancing global context modeling with fine detail recovery [16], [33]. Recent studies have also explored spatial–frequency integration to improve representation learning in RSI. For instance, MFVNet [34] employs multiview feature fusion to enhance representation diversity, and REST [35] focuses on holistic spatial modeling through region-level interactions. These methods mainly improve feature learning within the spatial domain. Different from these approaches, our work investigates frequency-band-oriented sequence modeling, which establishes structured dependencies across frequency components. This provides a complementary perspective to existing spatial-domain strategies.

Nevertheless, these paradigms still operate predominantly on spatial tokens, and directional structures with multiscale organization are not always explicitly characterized in purely spatial representations. To address this limitation, we argue that shifting the perspective from the spatial domain to the frequency domain offers a more effective way to disentangle these structures. Consequently, we propose FBSNet, which integrates frequency-domain state space modeling with spatial backbones, allowing for the explicit characterization of directional structures that are often overlooked by purely spatial encoders.

B. Frequency-Domain Methods

Beyond purely spatial-domain representation learning, frequency-domain analysis provides an explicit mechanism to separate large-scale structural information from directional detail cues. Classical wavelet theory shows that multiresolution representations can decompose an image into components capturing global structures and components emphasizing edges and direction-dependent details [17], [36], [37]. In the discrete setting, a 2-D discrete wavelet transform (2D-DWT) produces one approximation component and several detail components, commonly denoted as $\{LL, LH, HL, HH\}$, where LL captures low-frequency structures and $\{LH, HL, HH\}$ encode directional high-frequency variations (e.g., horizontal, vertical, and diagonal details) [36], [37]. Such decomposition is particularly relevant to high-resolution RSI, where man-made objects (e.g., roads, rooftop edges, and regular farmlands) exhibit pronounced directional patterns and multiscale organization. Motivated by this property, recent RSI segmentation methods increasingly incorporate wavelet or frequency-domain features to better characterize directional high-frequency components that are critical for boundary delineation and structure consistency. A representative line of work integrates wavelet decomposition into encoder–decoder architectures to preserve high-frequency details during downsampling and recover sharp boundaries during decoding. For example, Azimi et al. [38] proposed Aerial LaneNet by enhancing a symmetric fully convolutional network with wavelet transform for lane-marking segmentation in aerial imagery. More recently, Yang et al. [18] introduced SFFNet, which applies the Haar wavelet transform to decompose features into one low-frequency component and three directional high-frequency components corresponding to horizontal, vertical, and diagonal variations, and then performs spatial–frequency feature fusion to strengthen detail recovery. Related designs, such as the wavelet transform feature enhancement (WFE) method by Li et al. [39], further explore multiscale wavelet feature enhancement for remote sensing segmentation through hierarchical decomposition and reinjection of high-frequency components into the segmentation backbone. In addition, Gao et al. [20] presented AFENet, which introduces adaptive frequency enhancement by dynamically separating and modulating high-/low-frequency components and selectively fusing spatial–frequency features for RSI semantic segmentation. Beyond the remote sensing domain, wavelet operators have also been explored as differentiable components and combined with Transformer-style representation learning in general vision models (e.g., WaveSNet [40] and Wave-ViT [41]).

Overall, existing frequency-aware methods demonstrate consistent gains in preserving directional details by explicitly introducing approximation and detail components into deep architectures. Nevertheless, most approaches rely on fixed wavelet bases and fuse frequency components through relatively lightweight interactions, leaving the structured and hierarchical relationships among frequency bands insufficiently modeled. In contrast, our approach introduces an LFBD to adaptively partition frequency bands, overcoming the rigidity of fixed bases. Furthermore, instead of simple fusion, we propose the FBSS module to explicitly model the hierarchical dependencies evolving from low-frequency contexts to high-frequency details via state space propagation.

C. SSM-Based Methods

SSMs have recently emerged as an efficient alternative to self-attention for long-range dependency modeling. To apply SSMs to vision tasks, images are typically serialized into sequences, making scan routes a key factor in determining

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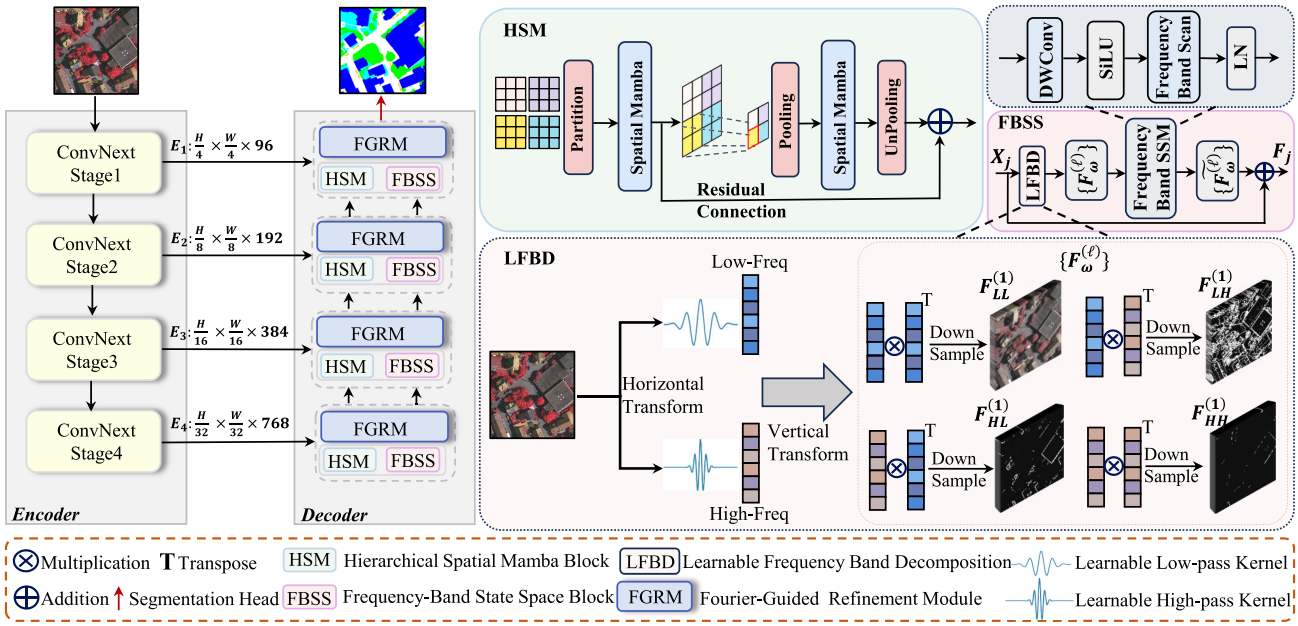


Fig. 2. Overall architecture of the proposed FBSNet. The network utilizes an FBSS module equipped with LFBD to model directional cross-band dependencies in the frequency domain. In the decoder, an HSM block is introduced to maintain global spatial consistency, while an FGRM leverages spectral priors to preserve high-frequency boundary details during progressive upsampling. This design ensures robust parsing of directional structures and precise boundary recovery.

how spatial information is propagated. Zhu et al. [22] proposed Vision Mamba with bidirectional state propagation to enhance visual representation learning while retaining linear complexity. To better preserve locality, Huang et al. [23] introduced LocalMamba by adopting windowed selective scan to maintain neighborhood consistency. In addition, hybrid backbones that combine SSM blocks with Transformer components have also been explored [42]. In remote sensing semantic segmentation, several recent works have started to tailor selective SSMs to high-resolution aerial and satellite imagery. Ma et al. [25] presented RS^{3}Mamba, a dual-branch framework that uses a Mamba-based branch to complement a convolutional branch with additional global context. Li et al. [43] proposed GLVMamba, which integrates global–local visual state space modeling and multiscale context aggregation for remote sensing image segmentation. Yang et al. [26] further explored spatial–channel stream modeling with an SSM formulation to establish dependencies along different feature planes with linear complexity. Moreover, Zhu et al. [44] conducted an experimental study on scanning strategies for Vision Mamba in remote sensing segmentation, highlighting the importance of scan design for dense prediction.

Overall, existing Mamba-based segmenters demonstrate the effectiveness of state space modeling in capturing long-range dependencies for remote sensing imagery. However, existing 2-D spatial-domain serialization strategies (e.g., VMamba's four-way cross-scan or RS3Mamba's spatial flattening) uniformly flatten pixels without accounting for intrinsic frequency semantics. Consequently, adjacent pixels of the same directional structure—such as elongated objects or sharp corners—can be separated in the 1-D sequence. This disruption of spatial proximity causes feature oversmoothing and fragments directional details, weakening boundary segmentation performance.

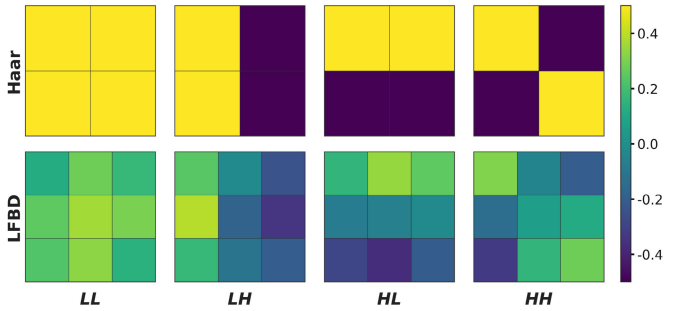


Fig. 3. Fixed Haar bases and learned LFBD kernels. The top row corresponds to the Haar bases of the LL, LH, HL, and HH sub-bands, and the bottom row corresponds to the kernels learned by LFBD. For visualization, the separable 1-D filters are shown in equivalent 2-D forms, and the learned kernels are averaged across channels. Relative to the predefined Haar patterns, the learned kernels present smoother spatial distributions while retaining the low-/high-frequency decomposition characteristics.

To bridge this gap, we shift the serialization axis from the spatial plane to the frequency domain. Unlike prior frequency-enhancement schemes, FBSS explicitly organizes sub-bands hierarchically and models cross-band dependencies. By serializing sub-bands into a direction-aligned sequence, our proposed FBSS module enables the SSM to capture the intrinsic cross-band dependencies, facilitating a more robust coupling of global layouts and local directional details.

III. Method

A. FBSNet Architecture

As shown in Fig. 2, we propose FBSNet, a frequency-aware state space network with an encoder–decoder paradigm for remote sensing semantic segmentation. The encoder adopts a ConvNeXt backbone [45] to extract a hierarchical feature pyramid, while the decoder progressively recovers high-resolution predictions by integrating frequency-band dependency modeling, spatial structural modeling, and Fourier-guided refinement

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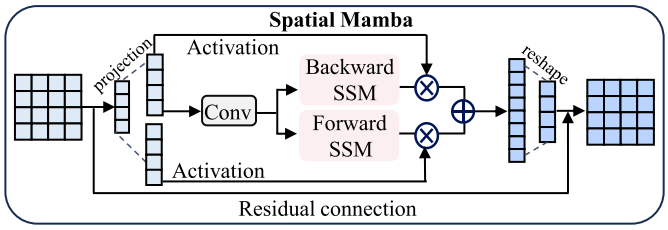


Fig. 4. Bidirectional Spatial Mamba operator used in HSM. The input feature is projected and processed by forward and backward selective scans (SSM) with residual connections, enabling efficient long-range dependency modeling along serialized spatial sequences.

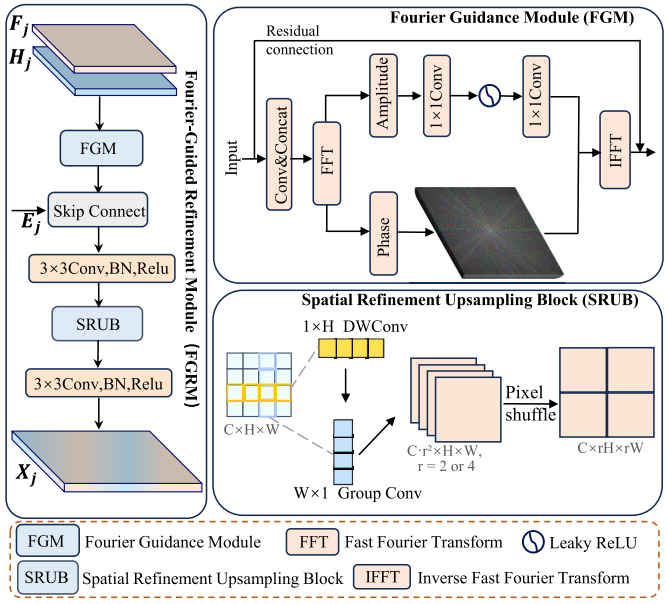


Fig. 5. Details of the FGRM. Given F_j and H_j, features are concatenated and transformed by FFT; the amplitude branch is enhanced while the phase is preserved to avoid spatial distortion; the refined feature is reconstructed by IFFT and fused with the encoder feature via SRUB.

(Fig. 2). The learned LFBD kernels and the bidirectional spatial Mamba operator are further illustrated in Figs. 3 and 4, respectively. At each decoding stage j, we incorporate the encoder feature $\mathbf{E}_j$ through skip connection in FGRM to refine the stage output $\mathbf{X}_j$ (Fig. 5). Specifically, FBSNet first applies an LFBD to produce multilevel sub-band features $\{F_\omega^{(l)}\}_{\omega\in\{\mathrm{LL,LH,HL,HH}\}}$ with explicit directional attributes. These sub-bands are fed into the FBSS module, which models structured directional cross-band dependencies by propagating information along the frequency-band axis to yield frequency-aware representations with enhanced structural consistency; the sub-band-aligned scan S_ω is used to serialize each $F_\omega^{(l)}$ into a bandwise sequence [Fig. 1(c)], with details given in Section III-B. In parallel, an HSM block performs spatial selective-scan modeling to enhance large-area geometric coherence and cross-scale structural stability, working in concert with the bandwise propagation in FBSS. Finally, a lightweight FGRM injects Fourier-domain cues during upsampling to improve boundary fidelity and alleviate high-frequency attenuation in progressive reconstruction. The refined feature is finally fed into a segmentation head to produce pixelwise class logits.

B. FBSS Module

The FBSS module constitutes the core engine for structured representation learning in our framework. As illustrated in Fig. 2, unlike conventional visual SSMs that serialize tokens purely in the spatial domain [46], FBSS shifts the propagation paradigm to the frequency-band axis. By redefining frequency sub-bands as the fundamental units of propagation, this module facilitates explicit cross-band interaction between global structural layouts and fine-grained directional details via linear-time sequence modeling [47].

To construct a task-adaptive frequency space, FBSS first employs LFBD to decompose the input feature $X^{(l)} \in \mathbb{R}^{C\times H\times W}$. Inspired by classical multiresolution analysis [17], we implement the decomposition operator $\mathcal{D}(\cdot)$ via separable learnable filtering

$$\left\{F_{\mathrm{LL}}^{(l)}, F_{\mathrm{LH}}^{(l)}, F_{\mathrm{HL}}^{(l)}, F_{\mathrm{HH}}^{(l)}\right\} = \mathcal{D}\left(X^{(l)}\right) \tag{1}$$

where the sub-bands are generated by depthwise 1-D low-/high-pass filtering along horizontal and vertical axes

$$\begin{aligned} F_{\mathrm{LL}}^{(l)} &= \phi_L^h\left(\phi_L^v\left(X^{(l)}\right)\right), \quad F_{\mathrm{LH}}^{(l)} = \phi_H^h\left(\phi_L^v\left(X^{(l)}\right)\right) \\ F_{\mathrm{HL}}^{(l)} &= \phi_L^h\left(\phi_H^v\left(X^{(l)}\right)\right), \quad F_{\mathrm{HH}}^{(l)} = \phi_H^h\left(\phi_H^v\left(X^{(l)}\right)\right). \end{aligned} \tag{2}$$

In this module, $\phi_{L/H}^{h/v}$ denote the learnable filters along the horizontal and vertical directions, implemented as depthwise 1-D convolutions. Specifically, each channel independently learns its unique kernels to capture channel-specific directional cues. The filters are initialized via the Kaiming normal strategy [48] with a default size of $K = 3$, providing a minimal yet efficient functional unit for frequency modeling. Unlike fixed wavelet bases (e.g., Haar) [18], this data-driven design allows the filters to autonomously converge toward task-adaptive waveforms that better represent anisotropic remote sensing textures, such as sharp building boundaries. As illustrated in Fig. 3, these learned kernels exhibit smoother spatial distributions and stronger adaptability compared to predefined Haar patterns. Consequently, F_{LL} captures the coarse structural layout, while $\{F_{\mathrm{LH}}, F_{\mathrm{HL}}, F_{\mathrm{HH}}\}$ explicitly encode high-frequency details with predominant horizontal, vertical, and diagonal directionality, respectively [17].

To effectively model the dependencies within these directional sub-bands, we propose a semantic-aligned serialization strategy [Fig. 1(c)]. Standard raster scanning often disrupts the geometric continuity of directional features. To address this, we explicitly align the scanning axes with the intrinsic semantics of each sub-band [44]: horizontal scanning is applied to F_{LH} to trace lateral boundaries, vertical scanning to F_{HL} to follow longitudinal structures, and diagonal scanning to F_{HH}. Meanwhile, F_{LL} adopts a raster scan to establish a stable global context. This axis-aware strategy is consistent with the advanced scanning taxonomy in visual Mmba, which utilizes bidirectional traversals to handle the noncausal nature of image sequences [22], [24], [49].

Formally, for each band $\omega \in \{\mathrm{LL,LH,HL,HH}\}$ at level L, we define a band-specific bidirectional scan-expand operator

$$s_\omega^{(L)} = \mathcal{S}_\omega\left(F_\omega^{(\ell)}\right) = \left[S_\omega^{\rightarrow}\left(F_\omega^{(\ell)}\right) \; ; \; S_\omega^{\leftarrow}\left(F_\omega^{(\ell)}\right)\right] \tag{3}$$

where $S_\omega^{\rightarrow/\leftarrow}$ denote forward/backward traversals along the designated axis, and $[\cdot;\cdot]$ denotes concatenation.

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Following serialization, FBSS constructs an ordered sequence S_j to facilitate coarse-to-fine information flow

$$S_j = \text{Concat}\left(s_{\text{LL}}^{(L)},\ s_{\text{LH}}^{(L)},\ s_{\text{HL}}^{(L)},\ s_{\text{HH}}^{(L)}\right) \tag{4}$$

where low-frequency structure is propagated first to guide the aggregation of high-frequency details, consistent with the workflow in Fig. 2(c). We then apply an SSM operator to model cross-band dependencies

$$\widetilde{S}_j = \text{SSM}_\Theta\left(S_j\right). \tag{5}$$

This mechanism enables selective information exchange among frequency bands with linear computational complexity [47]. Through this process, the global context from F_{LL} suppresses noise in high-frequency bands, while directional details refine object boundaries.

Finally, the enhanced sequences are reconstructed to the spatial domain via inverse scanning

$$\tilde{F}_\omega^{(\ell)} = \mathcal{S}_\omega^{-1}\left(\tilde{s}_\omega^{(L)}\right), \quad \omega \in \{\text{LL}, \text{LH}, \text{HL}, \text{HH}\}. \tag{6}$$

With a residual connection, the output feature is obtained as follows:

$$F_j = X_j + \text{Fuse}\left(\left\{\tilde{F}_\omega^{(\ell)}\right\}_\omega\right) \tag{7}$$

where Fuse$(\cdot)$ denotes feature fusion implemented by concatenating $\{\tilde{F}_\omega^{(\ell)}\}_{\omega\in\{LL,LH,HL,HH\}}$ along the channel dimension, followed by a convolution layer to mix channels. By explicitly coupling LFBD with direction-aligned serialization and state space propagation, FBSS ensures a structured and frequency-consistent representation for the subsequent decoding stages.

C. HSM Block

Achieving a balanced representation requires coordinating cross-band interactions with spatial context modeling. Therefore, we introduce the HSM block to enforce large-area geometric continuity alongside frequency analysis. As integrated into the decoder architecture shown in Fig. 2, HSM adopts a hierarchical propagation strategy comprising a region-level stage for local-to-mid-range continuity and a pooled stage for efficient global interaction.

The fundamental unit of HSM is the bidirectional spatial state space operator, denoted as $\mathcal{M}(\cdot)$. As detailed in Fig. 4, this operator follows the efficient design of Vision Mamba [22], performing linear projection, convolution, and gated state space modeling. Crucially, to capture noncausal visual dependencies, it adopts the bidirectional spatial serialization strategy illustrated in Fig. 2(a), which flattens 2-D features into sequences along forward and backward scanning routes. Mathematically, for an input token sequence $\mathbf{x}$, the operator is formulated as follows:

$$\begin{aligned}\left[\mathbf{u}, \mathbf{g}\right] &= \mathbf{x}\mathbf{W}_{\text{in}}, \quad \mathbf{z} = \sigma\left(\text{Conv}\left(\mathbf{u}\right)\right) \cdot \sigma\left(\mathbf{g}\right) \\ \mathcal{M}\left(\mathbf{x}\right) &= \text{SSM}\left(\mathbf{z}\right) \cdot \mathbf{W}_{\text{out}} + \mathbf{x}\end{aligned} \tag{8}$$

where SSM$(\cdot)$ aggregates the states from the bidirectional paths defined with linear complexity.

Based on this operator, HSM processes the input feature X_j in a coarse-to-fine manner. First, to enhance local structural continuity, we employ a nonoverlapping partition strategy and apply the operator to each local region

$$R_j = \mathcal{M}_{\text{region}}\left(\text{Partition}\left(X_j\right)\right). \tag{9}$$

Subsequently, to expand the receptive field to the global scale, the region-refined features are pooled into compact tokens. A second operator propagates states across these global representations, followed by unpooling and residual fusion

$$H_j = X_j + \text{Unpool}\left(\mathcal{M}_{\text{global}}\left(\text{Pool}\left(R_j\right)\right)\right). \tag{10}$$

By cascading regionwise and pooled global propagation, HSM effectively stabilizes long-range spatial structures that might be fragmented by bandwise processing.

D. Fourier-Guided Refinement Module

While the FBSS module effectively models cross-band dependencies, the high-frequency cues essential for boundary delineation are inevitably attenuated during the progressive upsampling process. To mitigate this, we propose the FGRM, which integrates global frequency-domain priors with directional spatial refinement to ensure structure-preserving reconstruction. As illustrated in Fig. 5, FGRM takes the fused decoder feature U_j and the encoder skip connection E_j as input, operating in two coordination stages.

The design of the Fourier guidance module (FGM) is grounded in the spectral property that the phase component preserves high-level structural semantics (e.g., object shapes), while the amplitude component captures low-level statistics and texture intensities [20]. Inspired by recent frequency-aware advancements in remote sensing, we aim to enhance feature intensity without distorting the spatial layout. Specifically, we transform the intermediate feature U_j into the spectral domain via 2-D FFT

$$\mathcal{F}\left(U_j\right) = \mathcal{A}_j \cdot e^{i\mathcal{P}_j} \tag{11}$$

where $\mathcal{A}_j$ and $\mathcal{P}_j$ denote the amplitude and phase spectrums, respectively. We apply a learnable modulation function $\Phi(\cdot)$ solely to the amplitude component $\mathcal{A}_j$ to adaptively recalibrate the feature response, while keeping the phase $\mathcal{P}_j$ fixed to preserve geometric integrity. The refined spectrum is then reconstructed via inverse FFT (IFFT)

$$G_j = \text{IFFT}\left(\Phi\left(\mathcal{A}_j\right) \cdot e^{i\mathcal{P}_j}\right) \tag{12}$$

where $\Phi(\cdot)$ is parameterized by channelwise 1×1 convolutions and LeakyReLU activation. Specifically, LeakyReLU is adopted in the modulation function to avoid the hard truncation introduced by the standard ReLU, thereby allowing negative responses to be partially preserved during amplitude recalibration. Its piecewise linear characteristic is also suitable for maintaining subtle variations in the frequency response, which is beneficial for the refinement of structural details during the inverse transform. The generated Fourier-guided feature G_j is subsequently concatenated with E_j for fusion.

To restore spatial resolution while preserving the anisotropic characteristics of man-made structures, we design the spatial refinement upsampling block (SRUB). As illustrated in Fig. 5, SRUB adopts a lightweight factorized modeling strategy. Specifically, after resolution recovery via PixelShuffle [50],

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the features are processed by two parallel branches to capture long-range dependencies: 1) a $1 \times H$ depthwise convolution (denoted as DWConv in Fig. 5) to aggregate vertical contextual information and 2) a $W \times 1$ group convolution (Group Conv) to harvest horizontal structural cues. This hybrid design, combining the efficiency of DWConv and the channel-mixing capability of group conv, effectively expands the receptive field along spatial axes to prevent semantic ambiguity in large-scale patterns with linear complexity [26]. Finally, the outputs are fused to generate the refined feature X_j.

By incorporating Fourier-domain amplitude–phase guidance into spatial refinement, FGRM strengthens the recovery of high-frequency boundaries and directional details that are critical for dense prediction. Notably, FGRM is designed to complement the proposed FBSS from a different perspective: FBSS focuses on modeling cross-band dependencies through frequency-band propagation, while FGRM enforces frequency-consistent reconstruction during decoding. Together, they form a coherent frequency-aware decoding scheme that preserves global structural layouts and improves the delineation of fine-grained object boundaries in remote sensing segmentation.

IV. Experiments

A. Experimental Settings

1) Datasets: To comprehensively evaluate the effectiveness and robustness of FBSNet, we conduct experiments on three widely recognized remote sensing semantic segmentation benchmarks: the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the LoveDA dataset [51], [52]. The detailed configurations are as follows.

ISPRS Vaihingen: The Vaihingen dataset consists of 33 true orthophoto (TOP) tiles acquired over a small village in Germany, with an average size of 2494 × 2064 pixels and a ground sampling distance (GSD) of 9 cm. Each image contains three spectral bands: near-infrared (NIR), red (R), and green (G). The dataset is annotated with six land-cover categories: impervious surface, building, low vegetation, tree, car, and clutter. Following the standard partition protocol used in previous works [18], we select 16 tiles for training (IDs: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37) and use the remaining 17 tiles for testing. Due to the limited number of large tiles, we crop them into 1024×1024 patches during training. Note that the clutter category is excluded from quantitative calculations due to its sparse distribution, following common practice.

ISPRS Potsdam: The Potsdam dataset contains 38 high-resolution TOP tiles (6000 × 6000 pixels) with a GSD of 5 cm, captured over a historic city. It provides four spectral bands: NIR, red, green, and blue. The semantic categories are consistent with the Vaihingen dataset. Compared to Vaihingen, Potsdam presents more complex urban scenes with finer details and high intraclass variation. Adhering to the official benchmark split, we utilize 24 tiles for training (IDs: 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_11, and 7_12) and the remaining 14 tiles for testing. Similarly, the large tiles are cropped into 1024×1024 patches for training and inference.

LoveDA: LoveDA is a large-scale dataset designed for domain adaptation and land-cover mapping, covering 5987 images with a resolution of 1024 × 1024 pixels and a GSD of 0.3 m. It encompasses diverse scenes from urban (Nanjing and Changzhou) and rural (Wuhan) areas, posing significant challenges due to multiscale objects and complex background textures. The dataset is officially split into 2522 training images, 1669 validation images, and 1796 testing images. It includes seven categories: background, building, road, water, barren, forest, and agriculture. Unlike the ISPRS datasets, LoveDA involves evident domain shifts between urban and rural distributions, making it suitable for evaluating model generalization.

2) Implementation Details: To ensure the reproducibility of experiments and the fairness of comparison, all experiments in this article are conducted under a unified training and testing protocol.

All experiments are implemented with PyTorch 3.10 and conducted on a single NVIDIA GeForce RTX 3090 Ti GPU. We adopt the AdamW optimizer with an initial learning rate of 6×10^{-4} and a weight decay of 0.01, and employ a cosine annealing learning rate schedule for stable convergence. The training and validation batch sizes are both set to 8. In addition, a smaller learning rate is applied to the backbone, and the Lookahead strategy is adopted to improve training stability. For fair comparison and consistent computation, all datasets use a unified input resolution of 1024×1024. Specifically, the large tiles in Vaihingen and Potsdam are divided into subtiles of this size for training and inference, whereas LoveDA is directly used without random cropping. During training, we apply standard data augmentation including random scaling with factors $\{0.5, 0.75, 1.0, 1.25, 1.5\}$, random horizontal flipping, and random vertical flipping. We train the model for 105 epochs on Vaihingen and Potsdam, and for 45 epochs on LoveDA. During testing, we adopt test-time augmentation (TTA) with rotations and flipping and fuse the predictions to improve robustness and segmentation accuracy.

Additionally, to evaluate model complexity, we calculate the floating-point operations (FLOPs), parameter count, and frames per second (FPS) when processing a $3 \times 512 \times 512$ input image, providing insights into computational efficiency and inference speed.

The loss function integrates cross-entropy (CE) with Dice loss to guide model optimization. This hybrid loss aims to leverage the strengths of both metrics, with CE focusing on pixelwise classification accuracy and Dice loss improving the model's performance on imbalanced datasets by emphasizing the correspondence between predicted and true regions. It is formulated as follows:

$$\mathcal{L}_{ce} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \tag{13}$$

$$\mathcal{L}_{dice} = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{\hat{y}_k^{(n)} y_k^{(n)}}{\hat{y}_k^{(n)} + y_k^{(n)}} \tag{14}$$

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{dice}. \tag{15}$$

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TABLE I

ABLATION STUDIES ON THE COMPONENTS OF FBSNET ON THE VAIHINGEN DATASET

Baseline	HSM	FBSS	FGRM	Params (M) ↓	FLOPs (G) ↓	mF1 (%) ↑	mIoU (%) ↑
✓				58.4	68.8	90.91	83.69
✓	✓			60.2	72.8	91.54	84.70
✓		✓		62.8	74.2	91.71	84.99
✓			✓	59.0	69.7	91.13	84.04
✓	✓	✓		65.0	79.1	91.97	85.43
✓	✓		✓	60.8	73.7	91.85	85.21
✓		✓	✓	63.4	75.1	91.91	85.33
✓	✓	✓	✓	**65.1**	**80.0**	**92.22**	**85.84**

The best results are indicated in bold black. ↑ indicates higher is better, ↓ indicates lower is better.

TABLE II

ABLATION ON THE INTEGRATION SCHEMES OF FBSS, HSM, AND FGRM ON THE VAIHINGEN DATASET

Config	Integration Scheme	mF1 (%)	mIoU (%)	OA (%)
A	FBSS ‖ HSM → FGRM	**92.22**	**85.84**	**93.99**
B	FBSS → HSM → FGRM	92.01	85.52	93.82
C	HSM → FBSS → FGRM	91.98	85.46	93.87
D	(FBSS+HSM) → FGRM	91.96	85.40	93.85
E	FGRM → (FBSS ‖ HSM)	91.77	85.11	93.78

A denotes the default parallel design, where FBSS and HSM are modeled in parallel before FGRM refinement. D denotes that the outputs of FBSS and HSM are first merged into a single branch before being fed into FGRM. In all settings, the compared variants only differ in the module integration scheme. The best results are highlighted in bold.

Here, N and K denote the number of samples and the number of classes, respectively. $y_k^{(n)}$ and $\hat{y}_k^{(n)}$ represent the one-hot encoding of the ground-truth semantic label and the corresponding softmax output of the network, where $n \in \{1, \ldots, N\}$. Specifically, $\hat{y}_k^{(n)}$ indicates the confidence that sample n belongs to class k.

3) Evaluation Metrics: To quantitatively assess the segmentation performance, we report three standard metrics: overall accuracy (OA), $F1$-score ($F1$), and mean Intersection-over-Union (mIoU). Let TP, FP, FN, and TN denote the number of true positives, false positives, false negatives, and true negatives, respectively.

OA measures the proportion of correctly classified pixels and is calculated as follows:

$$\mathrm{OA} = \frac{TP + TN}{TP + FP + TN + FN}. \tag{16}$$

The $F1$-score represents the harmonic mean of Precision and Recall, providing a balanced assessment of exactness and sensitivity. These metrics are formulated as follows:

$$\mathrm{Precision} = \frac{TP}{TP + FP}, \quad \mathrm{Recall} = \frac{TP}{TP + FN}$$
$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{17}$$

mIoU evaluates the overlap between predicted and ground-truth regions averaged across all classes, where TP_i, FP_i, and FN_i correspond to the metrics for class i. For a dataset with K categories, it is defined as follows:

$$\mathrm{mIoU} = \frac{1}{K} \sum_{i=1}^{K} \frac{TP_i}{TP_i + FP_i + FN_i}. \tag{18}$$

B. Ablation Experiment

To ensure experimental rigor and reproducibility, all ablation studies are conducted under the strictly unified protocol defined in Section IV-A, unless otherwise specified. Specifically, we vary only the target module configuration, while keeping other settings—including input resolution, optimizer, learning rate schedule, data augmentation strategies, and a hybrid loss that combines CE loss and Dice loss—constant.

1) Components of FBSNet: Table I presents the ablation results on the Vaihingen dataset. The integration of each proposed module yields consistent performance gains, validating the effectiveness of our architectural design. Incorporating FBSS alone boosts mIoU by 1.30% (83.69% → 84.99%), underscoring its critical role in capturing directional structures. In parallel, the inclusion of HSM improves mIoU by 1.01%, confirming the necessity of long-range spatial context for geometric continuity. When deployed jointly, these two modules elevate the mIoU to 85.43%, demonstrating strong spectral–spatial complementarity. Finally, integrating FGRM further enhances the mIoU to 85.84% by recovering high-frequency details with negligible computational cost, thereby achieving an optimal balance between accuracy and efficiency.

These quantitative findings are corroborated by the visual comparisons in Fig. 6. Most notably, the variant without FBSS [Fig. 6(b)] struggles to preserve the geometric integrity of structures. As highlighted by the red boxes, the boundaries of buildings appear jagged and structurally distorted, failing to maintain their intrinsic rectangular shapes and straight edges. Meanwhile, removing HSM [Fig. 6(c)] leads to noticeable fragmentation within homogeneous regions (e.g., impervious surfaces), manifesting as severe "salt-and-pepper" noise. In contrast, the absence of FGRM [Fig. 6(d)] results in coarse boundary delineation, where sharp corners and edges become blurred and ambiguous.

These observations confirm that FBSS, HSM, and FGRM are indispensable for disentangling directional targets, ensuring spatial coherence, and preserving boundary fidelity, respectively. (Further visualizations are provided in Figs. 7–10.)

As shown in Table II, the default parallel design (Config A) achieves the best performance (85.84% mIoU). Preserving parallel collaboration between FBSS and HSM outperforms serial or early coupling variants, as it optimally maintains their complementary spatial–frequency modeling. Furthermore, placing FGRM early (Config E) yields the worst result, confirming its crucial role for late-stage boundary refinement during decoding.

2) Impact of FBSS Module: The FBSS module serves as the core engine for capturing cross-band dependencies, explicitly enhancing directional boundary structures. Quantitatively, as detailed in Table I, incorporating FBSS into the Baseline yields substantial gains, boosting mIoU by 1.30 % (83.69% → 84.99%) and mF1 by 0.80%. Conversely,

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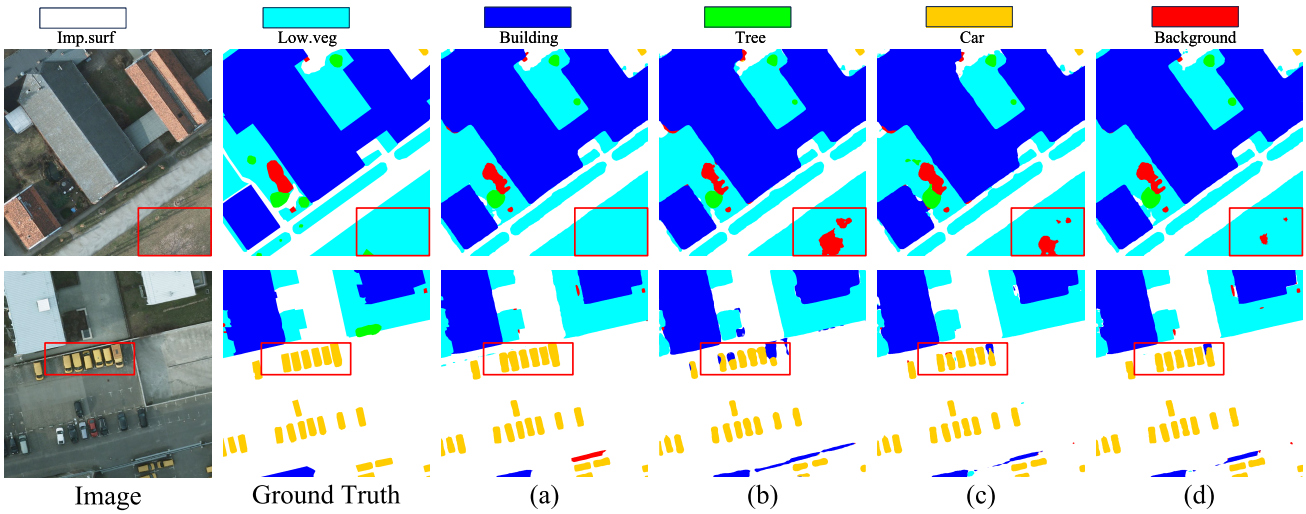


Fig. 6. Visualization comparison of FBSNet on the Potsdam dataset and its variants by removing individual modules. (a) FBSNet (full model). (b) FBSNet without the FBSS module. (c) FBSNet without the HSM module. (d) FBSNet without the FGRM module. The red boxes highlight regions with noticeable differences.

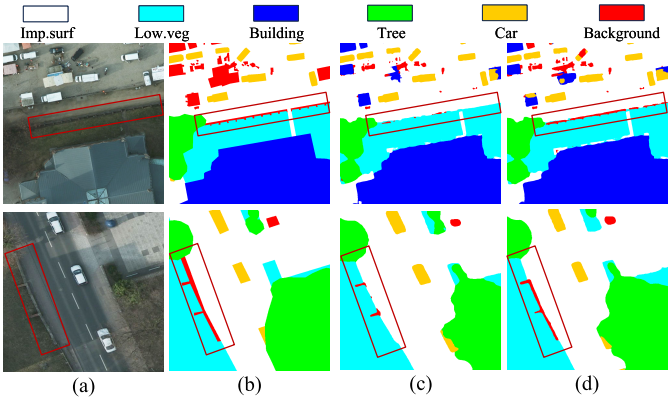


Fig. 7. Qualitative results for the FBSS-related visualization on the Potsdam dataset. (a) Input image. (b) Ground truth. (c) Baseline. (d) Baseline+FBSS. The red boxes highlight representative regions where FBSS improves the connectivity of linear structures.

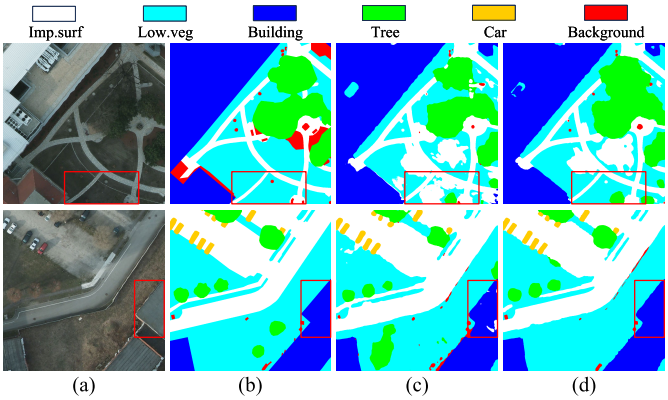


Fig. 9. Qualitative results for the HSM-related visualization on the Potsdam dataset. (a) Input image. (b) Ground truth. (c) Baseline. (d) Baseline+HSM. The red boxes highlight representative regions where HSM significantly reduces fragmentation and improves spatial continuity.

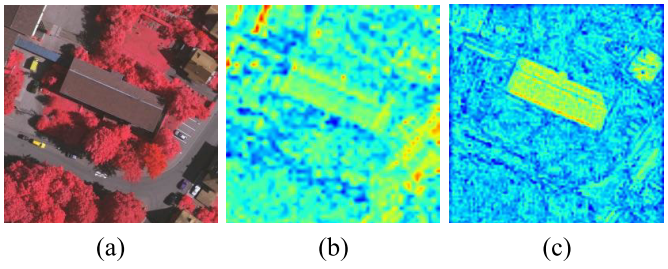


Fig. 8. Visualization of feature transformation through the proposed FBSS module. (a) Input image. (b) Feature responses input to the FBSS module, showing scattered activations and notable background noise. (c) Feature responses output by the FBSS module, where activations are highly concentrated on directional structures (e.g., buildings).

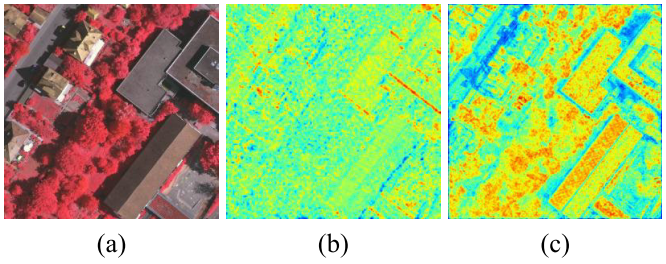


Fig. 10. Visualization of feature transformation through the HSM module. (a) Input image. (b) Feature responses input to the HSM module, exhibiting fragmented activation patterns and weak semantic consistency within large objects. (c) Feature responses output by the HSM module, showing holistic and uniform activations that effectively cover the target regions, demonstrating the module's ability to model long-range spatial dependencies.

removing FBSS leads to a clear degradation, confirming its role as a cornerstone for segmentation accuracy.

Qualitatively, visual comparisons in Fig. 7 demonstrate the module's efficacy in geometry preservation. Compared to the baseline [Fig. 7(c)], which often struggles with ambiguous boundaries, the FBSS-enhanced model [Fig. 7(d)] effectively restores fine-grained linear structures (e.g., boundaries of narrow roads and building edges). Unlike large-scale structures, these thin, orientation-sensitive details are highly prone to

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TABLE III

ABLATION STUDIES OF DIFFERENT LFBD CONFIGURATIONS IN FBSS ON THE ISPRS VAIHINGEN DATASET

Setting	Params (M)	FLOPs (G)	mF1 (%)	mIoU (%)	OA (%)
(a) LFBD decomposition level					
$L = 2$	64.3	78.9	92.16	85.75	93.95
$L = 3$ (Default)	65.1	80.0	**92.22**	**85.84**	**93.99**
$L = 4$	66.7	82.8	92.08	85.61	93.81
$L = 5$	68.0	84.5	91.84	85.20	93.71
$L = 3 \rightarrow$ Haar	64.7	79.4	91.86	85.25	93.77
(b) LFBD kernel size					
$K = 3$ (Default)	65.1	80.0	**92.22**	**85.84**	**93.99**
$K = 5$	65.7	80.3	92.06	85.60	93.90
$K = 7$	66.1	80.6	91.89	85.30	93.80

(a) studies the decomposition level L in LFBD, where "$L = 3 \rightarrow$Haar" denotes replacing the learnable LFBD with a fixed Haar wavelet decomposition. (b) studies the kernel size K in LFBD under the same initialization setting. The best results are highlighted in bold.

TABLE IV

ABLATION STUDIES OF DIFFERENT FBSS CONFIGURATIONS ON THE VAIHINGEN DATASET

Config	Setting	Params (M)	mF1 (%)	mIoU (%)
(a) FBSS: Number of Sub-bands				
2-band	{LL, All-H}	64.4	91.93	85.38
3-band	{LL, LH/HL, HH}	64.8	92.08	85.62
4-band (Ours)	{LL, LH, HL, HH}	65.1	**92.22**	**85.84**
(b) FBSS: Serialization Order				
Low-to-high (Ours)	{LL, LH, HL, HH}	65.1	**92.22**	**85.84**
High-to-low	{HH, HL, LH, LL}	65.1	91.89	85.28
Mixed-1	{LL, HH, LH, HL}	65.1	92.06	85.59
Mixed-2	{LH, HL, HH, LL}	65.1	91.98	85.44

All-H denotes the merged high-frequency components LH, HL, and HH. LH/HL denotes the merged band formed by LH and HL. The best results are highlighted in bold.

TABLE V

ABLATION ON THE SCANNING STRATEGIES ACROSS DIFFERENT FREQUENCY BANDS IN THE FBSS MODULE ON THE VAIHINGEN DATASET

Config	Scan Direction				mF1 (%)	mIoU (%)	OA (%)
	LL	LH	HL	HH			
A	R	R	R	R	92.00	85.47	93.83
B	H	H	H	H	92.05	85.55	93.95
C	R	V	H	D	91.90	85.31	93.79
D	R	H	V	R	92.14	85.72	93.97
E (Ours)	**R**	**H**	**V**	**D**	**92.22**	**85.84**	**93.99**

Scanning key: **R**: Raster, **H**: Horizontal, **V**: Vertical, **D**: Diagonal. Configs A to D represent various sub-optimal scanning variants, while Config E is our proposed semantic-aligned strategy. The best results are indicated in bold black.

disconnection; however, FBSS successfully maintains their continuity by explicitly modeling their directional dependencies across frequency bands.

To intuitively verify this mechanism, we visualize the intermediate feature response heatmaps in Fig. 8. As shown in Fig. 8(b), the feature activations input to the FBSS module appears scattered and discontinuous, struggling to differentiate foreground targets from background noise. In contrast, the output representations [Fig. 8(c)] exhibit a significant qualitative improvement, where activations become highly concentrated along the intrinsic structural directions of man-made buildings. This transformation corroborates that the proposed frequency-band serialization mechanism successfully propagates state information along the principal directions of targets, thereby maintaining structural integrity.

For the LFBD design (Table III), a decomposition level of $L = 3$ and a kernel size of $K = 3$ achieve the optimal performance (92.22% mF1, 85.84% mIoU). Deeper decompositions ($L \geq 4$) introduce fragmented high-frequency responses, while larger kernels ($K > 3$) tend to oversmooth delicate boundary textures. Furthermore, replacing the learnable LFBD with a fixed Haar wavelet degrades accuracy, confirming the necessity of task-adaptive frequency partitioning. Therefore, we adopt $L = 3$ and $K = 3$ as the default configuration.

Table IV further validates the internal design of FBSS. Our 4-band configuration ({LL, LH, HL, HH}) achieves the best performance (92.22% mF1, 85.84% mIoU), as merging high-frequency components (into 2-band or 3-band settings) weakens the explicit modeling of directional details. Additionally, the default low-to-high serialization order outperforms high-to-low or mixed variants. This confirms that propagating the low-frequency component first provides essential structural guidance for subsequent high-frequency aggregation, demonstrating the effectiveness of our coarse-to-fine strategy.

Finally, Table V empirically justifies the serialization scan strategy in FBSS. Our proposed semantic-aligned strategy (Config E) explicitly matches scanning axes to directional high-frequency cues, achieving the optimal performance (85.84% mIoU). In contrast, misaligning the scanning direction (Config C) actively disrupts structural continuity, while uniform raster or horizontal scans (Configs A and B) ignore intrinsic directional priors, yielding suboptimal results. This confirms that our semantic-aligned scanning is physically interpretable and essential for preserving complex anisotropic structures.

3) Impact of HSM Block: Designed as a spatial-domain counterpart to the frequency-aware FBSS, the HSM module is tailored to enforce structural continuity. While FBSS captures directional dependencies, HSM consolidates geometric coherence across large spatial extents, achieving a balanced representation. Quantitatively, as shown in Table I, adding HSM to the Baseline yields a consistent gain (mIoU: 83.69% $\rightarrow$ 84.70%). Conversely, removing it from the full model causes a noticeable drop (mIoU: 85.84% $\rightarrow$ 85.33%), confirming its reliability in maintaining structural stability.

Qualitatively, visual comparisons in Fig. 9 highlight the module's capability in mitigating intraclass fragmentation. Without HSM [Fig. 9(c)], predictions for homogeneous regions (e.g., large Low.veg areas and Building rooftops) often suffer from discontinuities and "salt-and-pepper" noise. In contrast, introducing HSM [Fig. 9(d)] effectively fills these semantic holes, generating spatially contiguous regions.

To further validate this capability, we visualize the internal feature transformation in Fig. 10. As observed in Fig. 10(b),

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TABLE VI
ABLATION ON THE PARTITION/POOLING CONFIGURATION IN HSM

Config	partition	pooling	mF1 (%)	mIoU (%)	OA (%)
Baseline	–	–	90.91	83.69	93.24
Baseline+HSM	2	1	90.87	83.65	93.11
Baseline+HSM	4	2	91.13	84.03	93.39
Baseline+HSM	8	4	**91.54**	**84.70**	**93.58**
Baseline+HSM	16	8	91.32	84.31	93.45
Baseline+HSM	32	16	91.17	84.11	93.40

The best results are indicated in bold black.

the feature maps input to HSM appears fragmented and patchy, confined to local texture details. In sharp contrast, the output representations [Fig. 10(c)] exhibit remarkable spatial enhancement, where activation patterns become holistic and uniform, effectively covering the entire extent of target regions. This transformation confirms that the hierarchical state space modeling mechanism successfully facilitates long-range information exchange, thereby repairing intraclass fragmentation and ensuring global geometric coherence.

Table VI analyzes the HSM partition size (local) and pooling size (global) to balance contextual coverage and detail preservation. Overly small partitions (e.g., 2) limit the local receptive field and disrupt spatial continuity, while excessively large partitions (e.g., 32) dilute fine local features with background noise. Consequently, the configuration (8, 4) achieves the optimal balance (84.70% mIoU) and is adopted as our default setting.

4) Impact of FGRM: The FGRM addresses the inevitable attenuation of high-frequency details (e.g., edges and corners) that occurs during decoder upsampling. By leveraging global spectral priors to rectify spatial features, it ensures structure-preserving reconstruction. Quantitatively, as shown in Table I, adding FGRM to the Baseline yields consistent gains (mIoU: 83.69% → 84.04%). More importantly, removing FGRM from the full model results in a clear performance drop (mIoU: 85.84% → 85.43%). Considering its negligible computational overhead (adding only 0.1M params as per Table I), FGRM proves to be a highly cost-effective component for refining dense predictions.

Qualitatively, comparisons in Fig. 11 demonstrate the module's role in boundary recovery. As highlighted by the red boxes, the Baseline [Fig. 11(c)] tends to produce blurred or jagged edges around man-made objects. In contrast, the FGRM-enhanced model [Fig. 11(d)] yields sharper boundaries and more stable contours. This visual evidence confirms the strategic complementarity between our modules: while FBSS enhances the extraction of directional features in the encoder, FGRM utilizes Fourier-domain amplitude modulation to ensure the fidelity of these features during the resolution recovery process.

5) Backbone Ablation: To verify the generalization capability of our framework, we conduct backbone ablation studies by replacing the encoder with various architectures, including CNN-based (ResNet50), Transformer-based (Swin-Tiny/Small), and ConvNet-based (ConvNeXt-Tiny/Small)

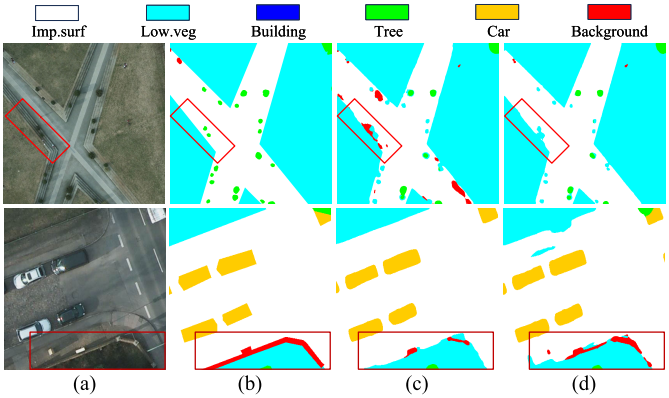


Fig. 11. Qualitative results for the FGRM-related visualization on the Potsdam dataset. (a) Input image. (b) Ground truth. (c) Baseline. (d) Baseline+FGRM. The red boxes highlight representative regions for comparison.

TABLE VII
BACKBONE ABLATION RESULTS

Backbone	Params (M)	FLOPs (G)	mF1 (%)	mIoU (%)
ResNet50	**33.4**	58.4	92.02	85.52
Swin-Tiny	43.0	60.2	91.95	85.40
Swin-Small	64.3	84.0	91.96	85.44
ConvNeXt-Tiny	43.3	**58.0**	92.08	85.61
ConvNeXt-Small	65.1	80.0	**92.22**	**85.84**

The best results are indicated in bold black.

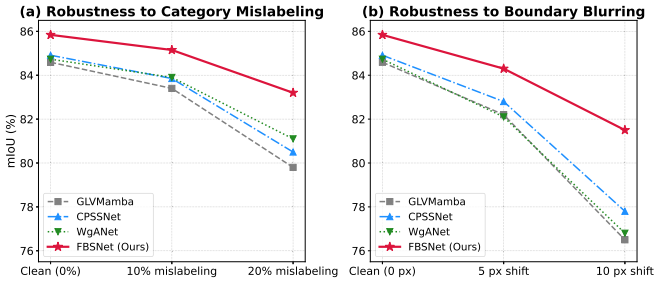


Fig. 12. Robustness evaluation under different label noise scenarios on the Vaihingen dataset. (a) Performance degradation under category mislabeling. (b) Performance degradation under boundary blurring, modeled by boundary shifts. FBSNet exhibits the strongest robustness, with the flattest degradation trend under both noise types.

backbones. As reported in Table VII, FBSNet achieves consistently competitive mF1 and mIoU scores across different backbones. This demonstrates that the performance gains are derived from the proposed frequency-aware modeling design rather than being dependent on a specific feature extractor. Considering the trade-off between accuracy and model complexity, ConvNeXt-Small achieves the optimal performance (mIoU = 85.84%).

C. Robustness to Label Noise

To evaluate robustness against typical annotation errors—category mislabeling (10% and 20% symmetric noise) and boundary blurring (five- and ten-pixel shifts) [53], [54]—we report model performance under noise in Fig. 12.

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TABLE VIII
QUANTITATIVE COMPARISONS ON THE VAIHINGEN DATASET (BOLD: BEST RESULTS)

Method	F1(%)					Evaluation index		
	Imp.surf	Building	Low veg.	Tree	Car	mF1(%)	mIoU(%)	OA(%)
U-Net [12]	95.79	92.77	81.91	88.69	82.40	88.31	79.51	91.51
SegFormer [15]	93.00	95.70	85.00	90.00	88.90	90.53	82.90	92.83
FT-UNetFormer [16]	96.83	95.79	85.04	90.28	87.22	91.03	83.88	93.42
EIGNet [55]	96.59	95.26	84.17	89.84	87.36	90.64	83.23	92.94
MANet [56]	96.74	95.46	84.61	89.72	88.74	91.05	83.89	93.07
CMTFNet [57]	96.61	95.15	84.93	89.87	86.36	90.58	83.12	93.04
CGGLNet [58]	96.87	95.53	84.25	90.26	88.31	91.04	83.90	93.28
SFFNet [18]	97.00	95.84	84.86	90.39	89.68	91.55	84.73	93.51
GLVMamba [43]	97.02	95.94	85.26	90.41	88.72	91.47	84.59	93.60
AFENet [20]	96.90	95.72	85.07	90.64	89.37	91.54	84.69	93.67
CPSSNet [26]	97.06	95.75	85.06	90.14	90.33	91.67	84.91	93.71
FSDENet [59]	96.97	95.96	85.27	90.59	88.63	91.48	84.61	93.56
WgANet [60]	97.03	96.03	85.38	90.60	88.73	91.56	84.73	93.64
FBSNet	**97.25**	**96.41**	**86.08**	**90.79**	**90.58**	**92.22**	**85.84**	**93.99**

While purely spatial-domain models (e.g., GLVMamba and CPSSNet) suffer steep mIoU drops under heavy boundary shifts due to misdirected spatial scanning, FBSNet maintains a significantly flatter degradation slope. This resilient performance confirms that our frequency-band serialization provides a stable structural anchor (via the low-frequency band) and enforces geometric consistency against localized label corruptions.

V. COMPARATIVE EXPERIMENTS

To comprehensively validate the effectiveness of the proposed method, we conduct comparative experiments with representative remote sensing semantic segmentation approaches, covering four major paradigms: CNN-based, Transformer-based, frequency-enhanced, and SSM-based methods. Specifically, the compared methods include the classical encoder–decoder baseline U-Net [12], the pure-Transformer segmentation framework SegFormer [15], and the hybrid architecture UNetFormer [16] (we adopt its stronger Swin-Transformer backbone variant in our implementation [14]). We also include attention-centric MANet [56] and the edge-guided model EIGNet [55], as well as CMTFNet [57] that fuses the strengths of CNNs and Transformers and CGGLNet [58] featuring category-guided global–local interaction. For the frequency-enhanced paradigm, we benchmark several recent frequency-aware and spatial–frequency hybrid methods. These include SFFNet [18], which explicitly performs spatial–frequency fusion via wavelet transform; AFENet [20], which adaptively separates and enhances high-/low-frequency components through FFT-based interaction; FSDENet [59], a recent dual-branch network utilizing Haar wavelets for detail enhancement alongside spatial features; and WgANet [60], which combines wavelet-guided attention with a spatial SSM to leverage both frequency and spatial domains. To reflect recent progress in long-range dependency modeling, we further compare with SSM-based methods, including CPSSNet [26] and GLVMamba [43], where the former introduces spatial and channel Mamba modules for linear-complexity remote dependency modeling, while the latter develops a global–local visual state space architecture tailored for remote sensing segmentation. For fair comparison, all experiments follow the official settings of the compared methods, and we implement them with the backbones specified in our comparison setting table.

A. Results on the Vaihingen Dataset

The quantitative results on the Vaihingen dataset are reported in Table VIII. FBSNet secures state-of-the-art performance across all global metrics, achieving an mF1 of 92.22% and an mIoU of 85.84%. Compared to the strongest competitor, CPSSNet, our method yields a solid mIoU improvement of 0.93%, demonstrating consistent superiority in segmentation quality. Moreover, when evaluated against the recently introduced frequency–spatial hybrid models, FSDENet and WgANet, FBSNet maintains a clear advantage. While FSDENet and WgANet achieve competitive mIoU scores of 84.61% and 84.73% respectively, they primarily rely on parallel spatial–frequency fusion or restrict SSMs to spatial tokens of the low-frequency band. In contrast, FBSNet dynamically serializes sub-bands, outperforming them by over 1.1% in mIoU.

Crucially, the performance gains are most pronounced in categories exhibiting strong directional geometric properties, aligning well with our frequency-aware design. For instance, the $F1$ scores of 97.25% on Imp.surf and 96.41% on Building indicate exceptional capability in delineating long-range directional structures, such as continuous roads and regular roof boundaries. Similarly, FBSNet achieves 90.58% $F1$ on Car, a representative category of small, man-made, and orientation-sensitive objects. This result confirms that our method effectively preserves high-frequency directional details, thereby minimizing missed detections. Furthermore, FBSNet maintains a competitive edge on Low veg. (86.08%) and Tree (90.79%), suggesting robust fine-grained discrimination even under complex textures and boundary ambiguities.

Qualitative comparisons in Fig. 13 further validate these metrics. The red boxes highlight challenging regions

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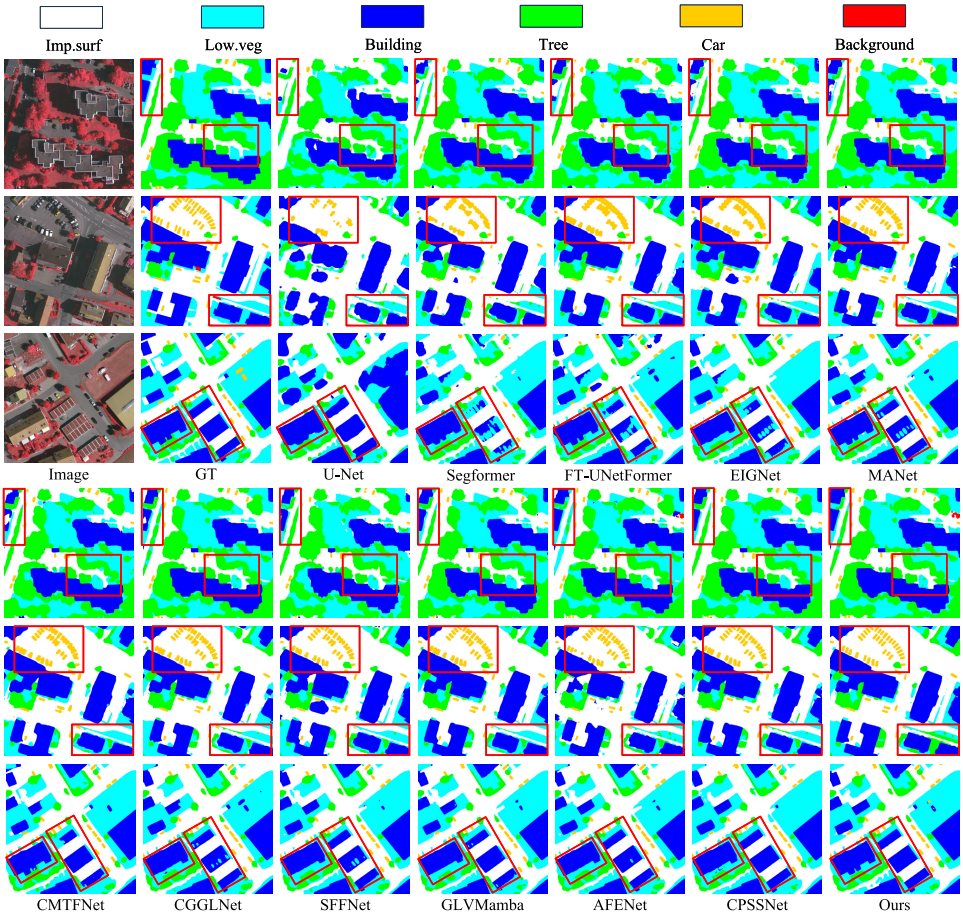


Fig. 13. Qualitative comparisons on the Vaihingen dataset (1024 × 1024). The red boxes highlight typical regions dominated by directional structures (e.g., roads/impervious surfaces and building edges) and small oriented objects (e.g., cars).

TABLE IX
QUANTITATIVE COMPARISONS ON THE POTSDAM DATASET (BOLD: BEST RESULTS)

Method	F1(%)					Evaluation index		
	Imp.surf	Building	Low veg.	Tree	Car	mF1(%)	mIoU(%)	OA(%)
U-Net [12]	92.25	94.09	86.27	87.34	95.71	91.13	83.92	89.38
SegFormer [15]	92.07	95.86	86.74	88.05	95.21	91.59	84.69	90.03
FT-UNetFormer [16]	92.68	96.24	87.06	88.79	95.68	92.09	85.56	90.67
EIGNet [55]	93.68	96.35	88.29	89.30	96.71	92.86	86.88	91.41
MANet [56]	93.77	96.44	87.63	89.16	96.40	92.68	86.58	91.42
CMTFNet [57]	93.86	96.15	88.13	89.13	96.63	92.78	86.73	91.49
CGGLNet [58]	94.14	97.06	87.57	89.03	96.64	92.89	86.97	91.52
SFFNet [18]	94.21	96.53	87.69	88.71	96.22	92.67	86.58	91.45
GLVMamba [43]	93.40	96.96	88.32	89.36	96.52	92.90	86.94	91.48
AFENet [20]	94.29	96.52	87.85	89.15	96.27	92.82	86.81	91.46
CPSSNet [26]	94.49	96.62	87.90	89.17	96.65	92.97	87.08	91.79
FSDENet [59]	94.40	96.72	87.40	88.78	96.32	92.72	86.68	91.48
WgANet [60]	93.64	97.13	87.99	89.25	96.33	92.87	86.90	91.50
FBSNet	**94.80**	**97.17**	**88.70**	**89.59**	**96.95**	**93.44**	**87.90**	**92.26**

dominated by directional structures (e.g., road/building edges) and small oriented objects. While competing methods often struggle with jagged boundaries, broken connectivity, or category adhesion, FBSNet yields cleaner, straight boundaries and maintains coherent structural continuity. For small vehicles, where baselines tend to merge instances into the impervious background, FBSNet preserves distinct object shapes with minimal scattered errors.

B. Results on the Potsdam Dataset

The quantitative comparisons on the Potsdam dataset are presented in Table IX. FBSNet achieves the best performance across all global metrics, recording 93.44% mF1, 87.90% mIoU, and 92.26% OA. Compared to the strongest baseline, CPSSNet [26], our method yields consistent gains of 0.47%, 0.82%, and 0.47% respectively. These margins substantiate the

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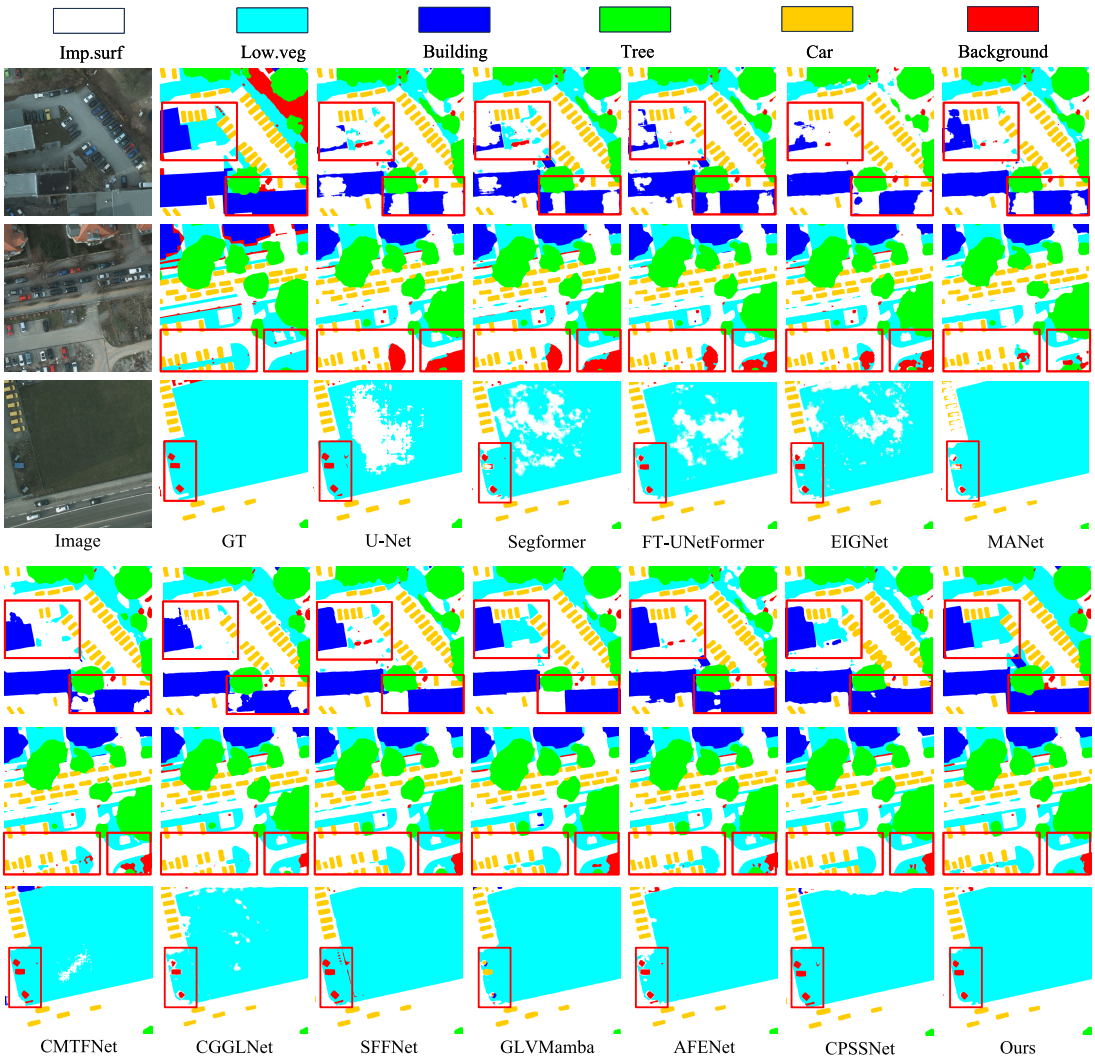


Fig. 14. Qualitative comparisons on the Potsdam dataset (1024 × 1024). The red boxes highlight representative regions with intraregion integrity issues (e.g., interior holes of buildings and broken/adhesive impervious blocks) and densely distributed small objects (cars), where structural continuity and small-object integrity are critical.

robustness of FBSNet in parsing the complex urban landscapes characterizing the Potsdam scene.

Specific classwise results further highlight our model's strengths. For the Car category, FBSNet achieves a remarkable 96.95%, confirming its ability to preserve high-frequency directional cues for small, orientation-sensitive objects. Furthermore, the top performance on Imp.surf (94.80%) and Building (97.17%) validates the effectiveness of the FBSS module. Since these categories largely comprise strip-shaped roads and regular roof edges, this result proves the model's capability in delineating large-scale directional man-made structures.

Qualitative comparisons are visualized in Fig. 14. The selected regions focus on challenges related to intraregion integrity and the separation of densely distributed small objects. As highlighted by the red boxes, competing methods frequently exhibit fragmented predictions within homogeneous surfaces or boundary adhesion between impervious grounds and buildings. In contrast, FBSNet generates spatially coherent regions with sharper, more continuous boundaries. For small vehicles, it maintains object shape integrity with fewer artifacts. These visual improvements align well with our architectural design: the FBSS module enhances anisotropic feature extraction for directional structures, the HSM module enforces global spatial consistency, and the FGRM further refines high-frequency details during upsampling to ensure boundary fidelity.

C. Results on the LoveDA Dataset

The split protocol of LoveDA differs significantly from that of Vaihingen and Potsdam: its training, validation, and test sets are divided by geographic locations, covering both urban and rural scenes simultaneously. Therefore, models are required to not only handle multiscale targets and inconsistent class distributions but also remain robust under cross-region distribution shifts.

The quantitative comparisons on the LoveDA dataset are presented in Table X. Under this challenging domain-shift-oriented setting, FBSNet achieves the best overall performance, reaching an mF1 of 70.60%, mIoU of 55.47%,

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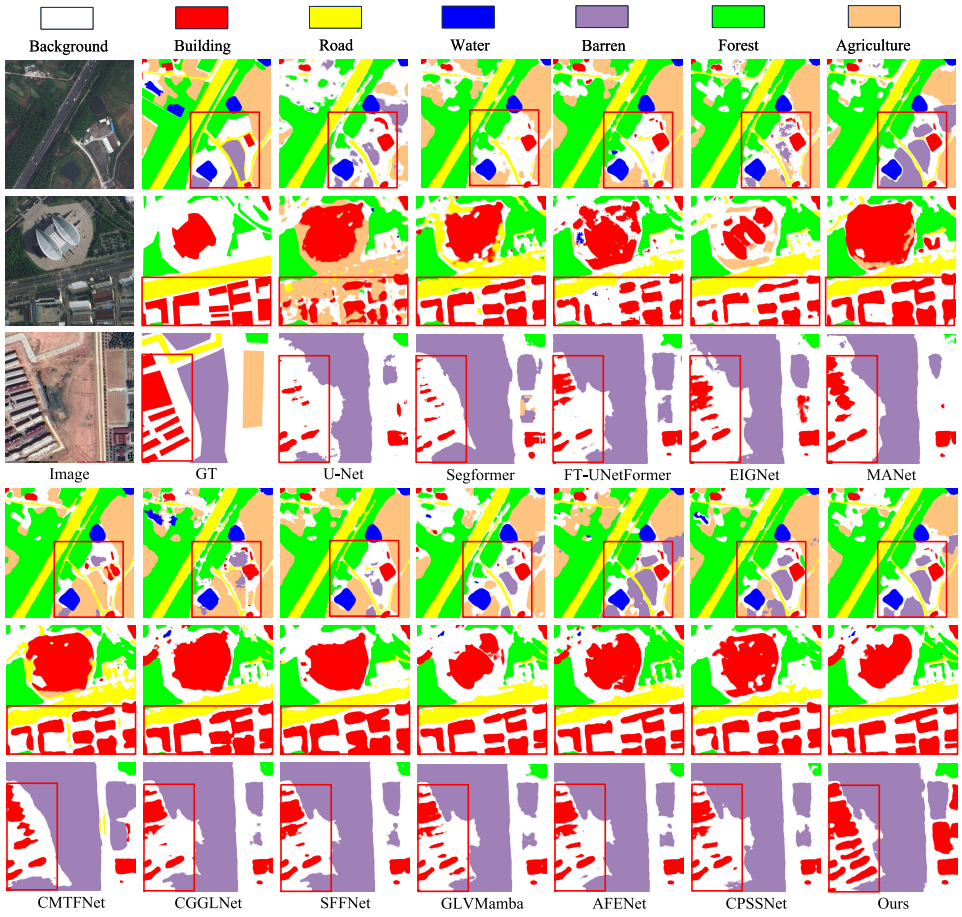


Fig. 15. Qualitative comparisons on the LoveDA dataset (1024×1024). Following the standard LoveDA visualization protocol (displaying two rural scenes and one urban scene), the red boxes highlight challenging regions, such as the interiors of agricultural blocks and the boundaries between barren land and farmland. While competing methods often produce holes or broken connectivity in these areas, FBSNet preserves the structural completeness of agricultural regions and the continuity of elongated roads. This demonstrates the robustness of our frequency-aware directional modeling in handling the complex urban–rural mixtures and cross-region disparities present in LoveDA.

and OA of 72.71%. Compared with the strongest baseline CPSSNet [26], our method improves the mIoU by 1.0%, demonstrating superior generalization capabilities for cross-domain land-cover mapping.

Crucially, our method secures the top performance in Building (65.85%), Road (57.25%), and Agriculture (60.88%). It is worth noting that these categories represent typical man-made or human-interfered targets exhibiting pronounced directional characteristics and geometric regularity. For instance, Agriculture in the LoveDA dataset often features large-scale continuous regions with stripe-like cultivation textures, while Roads present elongated directional structures. The significant gain in Agriculture (outperforming the second-best GLV-Mamba [43] by 3.88%) strongly validates the effectiveness of our proposed FBSS module, which explicitly models directional frequency-band dependencies to capture anisotropic textures, and the HSM module, which enforces large-scale spatial consistency. This combination effectively mitigates the fragmentation often observed in large agricultural plots and maintains the connectivity of road networks.

Qualitative comparisons are presented in Fig. 15, following the standard visualization setting (two rural scenes and one urban scene). As highlighted by the red boxes, several competing methods tend to produce holes or breaks inside large agricultural blocks or cause category adhesion near the boundaries between barren land and farmland. In contrast, FBSNet generates more complete predictions within agricultural regions with stable boundaries and preserves better continuity for elongated directional structures such as roads. These visual results align with the quantitative improvements, confirming that explicitly organizing directional cues across frequency bands significantly enhances robustness against the complex spatial heterogeneity found in urban–rural mixtures.

D. Complexity Analysis

We further compare different methods in terms of model complexity and inference efficiency. Table XI reports the number of parameters (Params), computational cost (FLOPs), and inference speed (FPS), together with the segmentation accuracy (mIoU). Overall, a clear trade-off can be observed among Params, FLOPs, and FPS. For instance, GLVMamba achieves the highest FPS (192.0) with very low FLOPs (7.7G), yet its mIoU (84.59%) is still lower than the best-performing method. AFENet is relatively lightweight in Params (20.2M) and FLOPs (25.6G), but its speed and accuracy remain limited (14.8 FPS and 84.69% mIoU). In contrast, our method

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TABLE X

QUANTITATIVE COMPARISONS ON THE LOVEDA DATASET (BOLD: BEST RESULTS)

Method	IoU(%)							Evaluation index		
	Background	Building	Road	Water	Barren	Forest	Agriculture	mF1(%)	mIoU(%)	OA(%)
U-Net [12]	50.50	56.72	47.80	52.63	30.43	40.66	45.69	62.90	46.35	65.14
SegFormer [15]	51.41	60.12	54.91	60.44	33.12	41.06	49.55	66.39	50.09	67.80
FT-UNetFormer [16]	52.39	58.29	52.40	62.85	**38.83**	38.40	48.89	66.49	50.29	67.96
EIGNet [55]	52.25	56.15	52.04	70.14	34.08	42.43	51.60	67.13	51.24	68.89
MANet [56]	53.31	57.73	52.50	66.40	37.39	38.31	53.62	67.29	51.33	69.45
CMTFNet [57]	50.65	63.91	55.34	55.75	37.31	**45.17**	55.42	67.97	51.94	70.18
CGGLNet [58]	54.33	65.45	54.48	69.00	31.74	38.90	52.10	67.78	52.30	70.35
SFFNet [18]	54.37	64.22	56.56	68.68	32.95	44.66	52.02	68.87	53.35	70.63
GLVMamba [43]	**55.48**	62.10	56.03	**72.26**	33.34	43.12	57.00	69.51	54.19	71.90
AFENet [20]	54.31	64.84	55.97	70.84	34.63	39.48	56.76	69.18	53.83	71.71
CPSSNet [26]	54.62	64.79	56.88	70.48	35.86	43.84	54.85	69.90	54.47	72.00
FSDENet [59]	54.35	64.78	57.02	70.13	33.52	43.41	55.67	69.15	54.16	71.89
WgANet [60]	55.09	65.44	55.54	70.49	35.80	42.25	55.98	69.55	54.35	71.94
FBSNet	54.16	**65.85**	**57.25**	71.24	34.74	44.18	**60.88**	**70.60**	**55.47**	**72.71**

TABLE XI

COMPLEXITY AND EFFICIENCY COMPARISON IN TERMS OF PARAMS (M), FLOPS (G), FPS, AND MIOU (%) ON THE VAIHINGEN DATASET

Method	Backbone	Params (M)↓	FLOPs (G)↓	FPS↑	mIoU (%)↑
U-Net [12]	ResNet-18	31.0	193.4	28.2	79.51
SegFormer [15]	MiT	84.6	110.2	37.4	82.90
FT-UNetFormer [16]	Swin-S	96.0	128.4	14.0	83.88
EIGNet [55]	ResNet-18	34.5	45.1	28.3	83.23
MANet [56]	ResNet-18	35.9	54.5	58.5	83.89
CMTFNet [57]	ResNet-50	30.1	34.6	12.2	83.12
CGGLNet [58]	ResNet-50	36.9	88.5	23.2	83.90
SFFNet [18]	ConvNeXt-T	34.2	52.1	34.7	84.73
GLVMamba [43]	ResNet-18	24.2	**7.7**	**192.0**	84.59
AFENet [20]	ResNet-18	**20.2**	25.6	14.8	84.69
CPSSNet [26]	ConvNeXt-T	31.9	28.8	47.3	84.91
FSDENet [59]	ConvNeXt-S	61.2	87.8	30.2	84.61
WgANet [60]	ResNet-18	117.8	52.0	8.9	84.73
FBSNet	ConvNeXt-S	65.1	80.0	19.1	**85.84**

The best results are indicated in bold black. ↑ indicates higher is better, ↓ indicates lower is better.

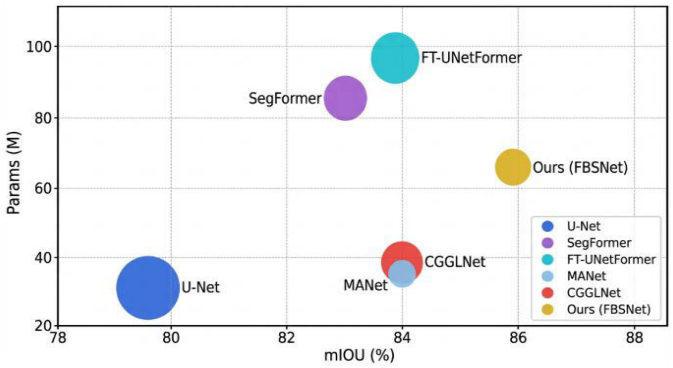


Fig. 16. Accuracy–complexity trade-off visualization on the Vaihingen dataset. The *x*-axis denotes mIoU, the *y*-axis denotes Params (M), and the bubble area represents FLOPs (G). Representative methods are selected for better visual clarity.

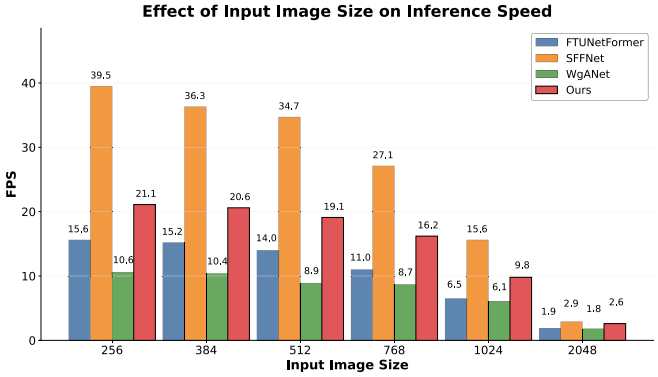


Fig. 17. Effect of input image size on inference speed (FPS). FBSNet exhibits more stable scalability across different input resolutions compared with representative baselines.

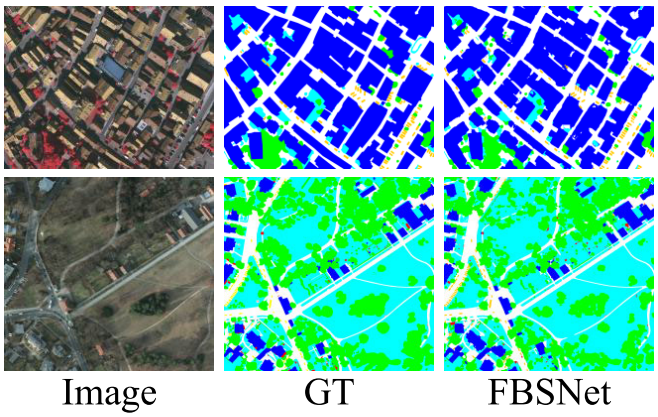


Fig. 18. Seamless patch-based inference on large-scale tiles (top: Vaihingen, bottom: Potsdam). FBSNet ensures structural continuity across the entire scene.

achieves the best mIoU (85.84%) while maintaining a reasonable complexity and efficiency (65.1M Params, 80.0G FLOPs, and 19.1 FPS), demonstrating a favorable accuracy–efficiency trade-off for practical land-cover mapping scenarios.

To provide a more intuitive view of the accuracy–complexity relationship, Fig. 16 presents a bubble-chart comparison on the Vaihingen dataset, where the *x*-axis denotes mIoU, the *y*-axis denotes Params, and the bubble area corresponds to FLOPs. It can be observed that our method achieves higher accuracy with a moderate parameter scale and avoids excessively high computational overhead, further validating the competitiveness of our design in terms of the accuracy–efficiency balance.

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To evaluate scalability for practical applications, we analyze the inference speed across input resolutions ranging from 256×256 to 2048×2048. As shown in Fig. 17, FBSNet maintains relatively stable efficiency at higher resolutions, demonstrating favorable scalability compared with representative baselines. Furthermore, we further validate FBSNet on large-scale remote sensing images using a sliding-window strategy. As visualized in Fig. 18, our model generates seamless and globally coherent predictions without visible stitching artifacts, demonstrating its practical viability for large-scale remote sensing mapping.

VI. Conclusion

In this article, we presented FBSNet, a novel state space framework that re-envisions the modeling of anisotropic and directionally organized structures in remote sensing imagery. Rather than treating these man-made targets solely as spatial patterns, we identified that their directionality inherently manifests as ordered multifrequency cues—ranging from global layouts to fine-grained-oriented details. To harness this property, we proposed the FBSS module grounded in an LFBD. By shifting the state-propagation axis from the image plane to a direction-aligned frequency sequence, our method explicitly captures cross-band dependencies, thereby sharpening orientation-sensitive features such as roads, building boundaries, and aligned vehicles. Coordinated with the HSM for global geometric coherence and the FGRM for high-frequency detail recovery, FBSNet ensures both structural connectivity and boundary fidelity. Extensive experiments on the Vaihingen, Potsdam, and LoveDA benchmarks demonstrate state-of-the-art performance with a favorable accuracy–efficiency trade-off. Ultimately, this work validates that organizing directional information along the frequency dimension offers a computationally efficient and structurally consistent alternative to purely spatial approaches, proving that explicit frequency-band dependency modeling is essential for parsing complex anisotropic scenes.

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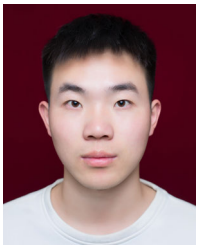
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